Sahaj Bhadja

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Summary

Software engineer with 2+ years building AI evaluation workflows, simulation pipelines, scalable backend services, and React applications using Python, TypeScript, PostgreSQL, cloud infrastructure, automated testing, and reliability-focused system design.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant AI services on PostgreSQL, Azure SQL, and Blob Storage for 500+ users, optimizing schemas and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI orchestration for 1,000+ monthly LLM jobs using explicit states, retries, evaluation checkpoints, and failure isolation, tripling throughput while strengthening production reliability and debugging visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows integrating GPT-4o, Claude, Qwen-VL, and OCR to evaluate 10,000+ daily submissions while maintaining stable production performance and 90%+ grading consistency.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch using multithreading, secure ingestion, and fault-tolerant execution, reducing model evaluation turnaround by 60% under peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer environment for autonomous-driving simulations, processing 100,000+ trajectory logs and improving long-horizon planning by 18% through reproducible data pipelines and systematic evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry and historical agent trajectories through multi-head attention, improving closed-loop simulation stability by 15% across diverse, complex real-world autonomous driving scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release reliability while validating integrations across Windows, Linux, and macOS environments.
- Optimized REST API and ETL components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex product requirements into maintainable real-time workflow domains.
- Automated Docker-based testing and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent production platform releases.

Skills

AI & Evaluation: Python, machine learning, deep learning, reinforcement learning, Decision Transformers, simulation environments, model evaluation, benchmarking, data quality, LLM orchestration, GPT-4o, Claude, Qwen-VL, OCR, prompt engineering, human feedback; **Applications & Data:** Java, C++, JavaScript, TypeScript, React.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, data modeling, ETL, data pipelines; **Cloud & Reliability:** AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, automated testing, debugging, monitoring, performance optimization, fault tolerance, distributed systems, system design, Agile/Scrum

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ simulated process combinations while reducing algorithm-debugging time by 20%.

CaptionCraft Live | *Python, PyTorch, Transformers, Flask, Gradio*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom transformer models on 10,000 images and 100,000 captions, improving evaluation accuracy from 81% to 96% through systematic model experimentation.

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built an AI data platform for 200+ users with secure ingestion, vector indexing, semantic retrieval, and collaborative workflows, maintaining 95% query accuracy across 50,000+ indexed documents.